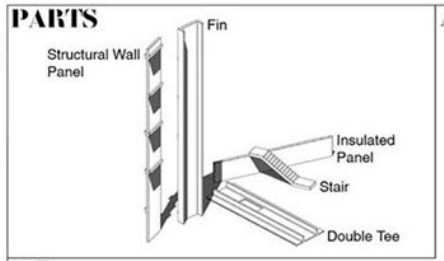
STRUCTURAL SYSTEMS 1 *ARC 2713*

CONCRETE BUILDING: PROJECT PARAMETERS & PROTOTYPES

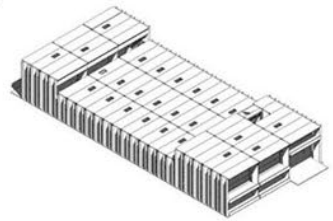
4 PRECAST PROTOTYPES

ALPHA TYPE A

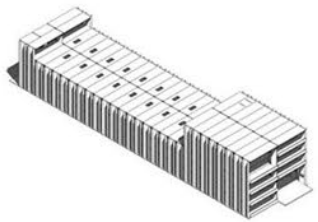
PARTS



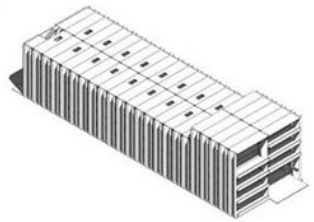
A.2



A.3

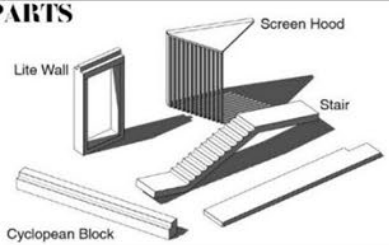


A.4

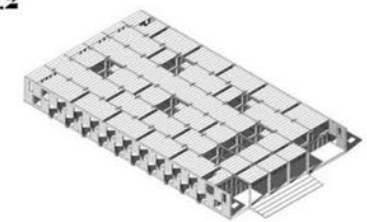


KILO TYPE K

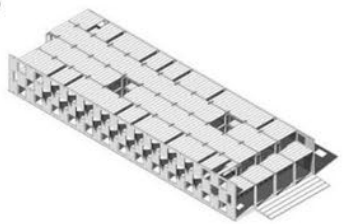
PARTS



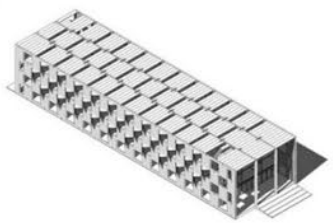
K.2



K.3

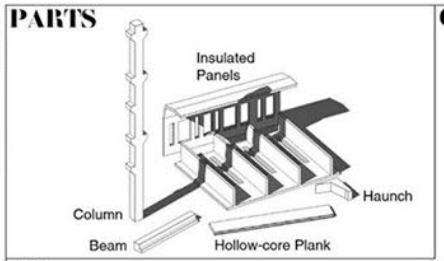


K.4

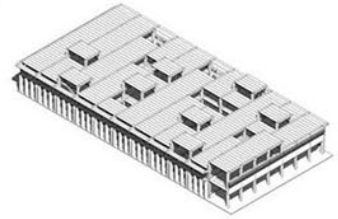


CHARLIE TYPE C

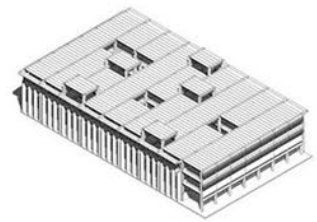
PARTS



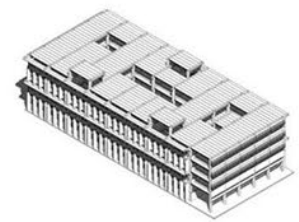
C.2



C.3

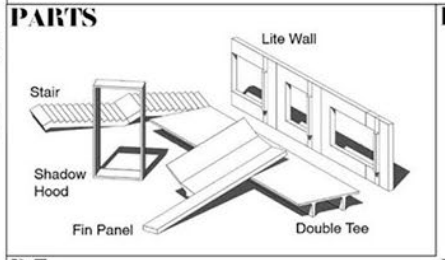


C.4

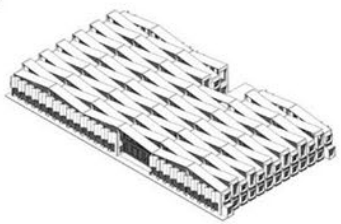


DELTA TYPE D

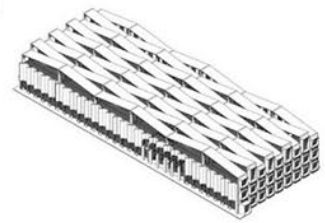
PARTS



D.2



D.3



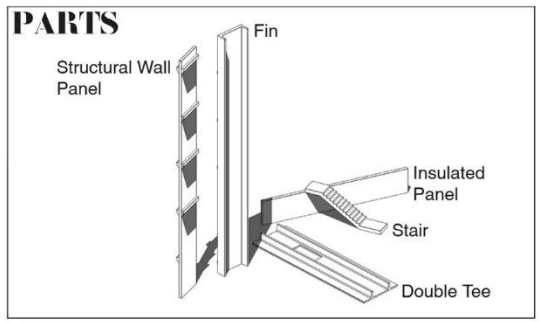
D.4

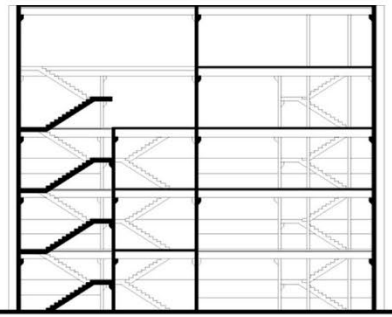


ALPHA
TYPE

A.3
VARIANT

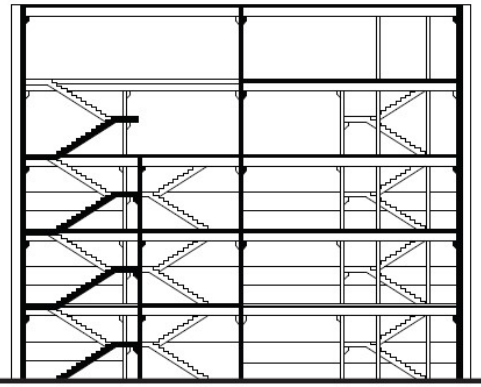




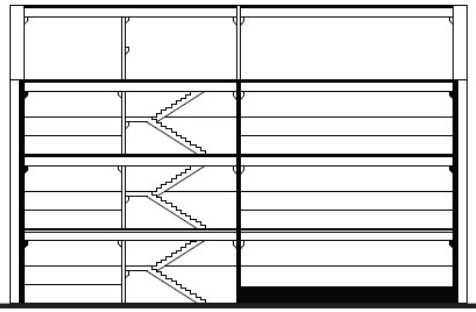




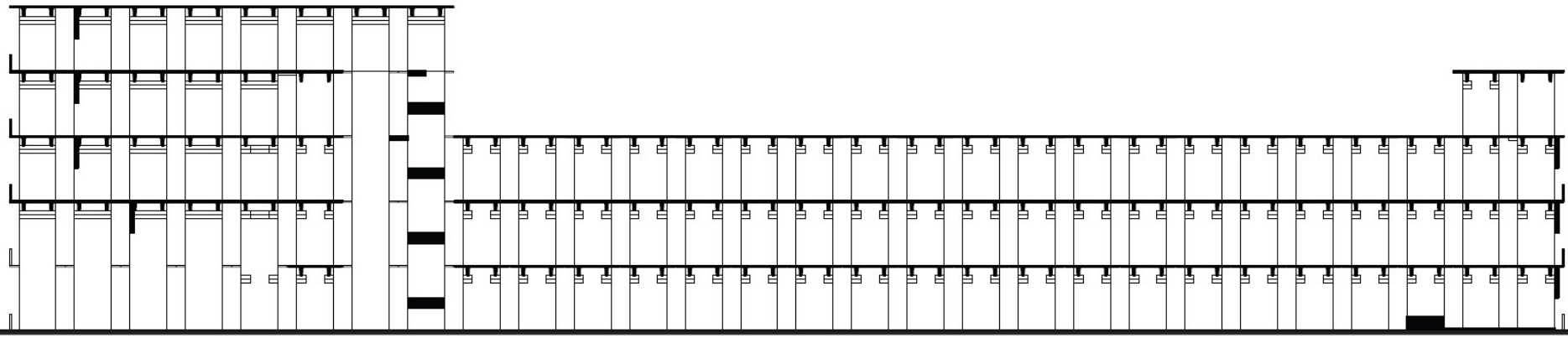
Section:
T1



Section:
T2



Section:
L1



ALPHA

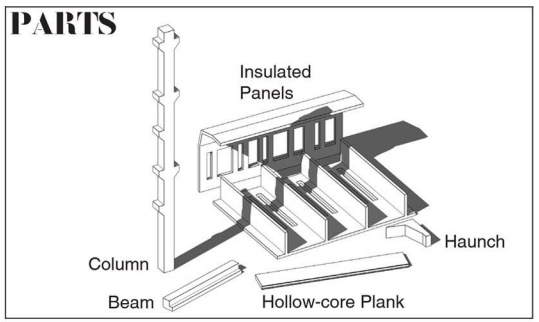
CHARLIE

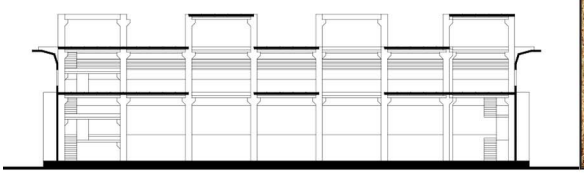
TYPE

C.2

VARIANT

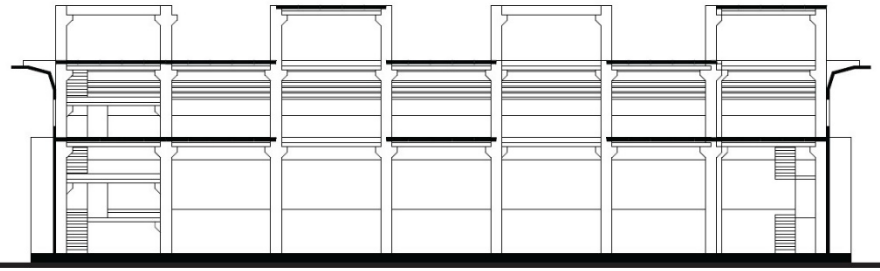




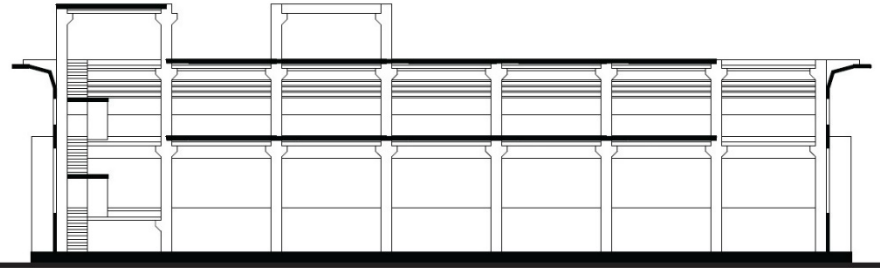




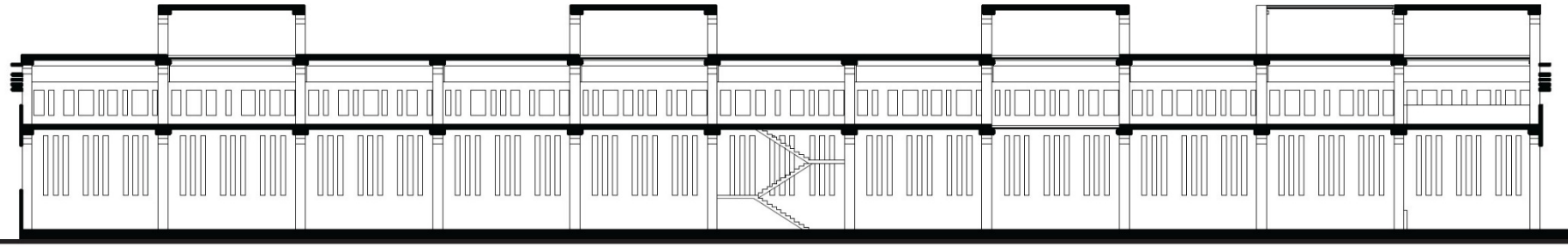
Section:
T1



Section:
T2



Section:
L1

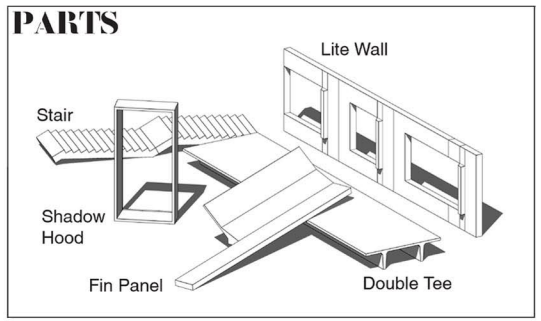


CHARLIE

DELTA
TYPE

D.3
VARIANT

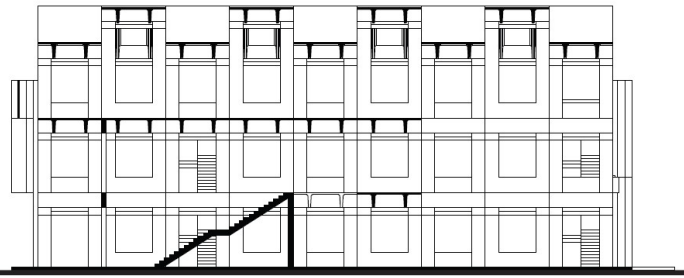




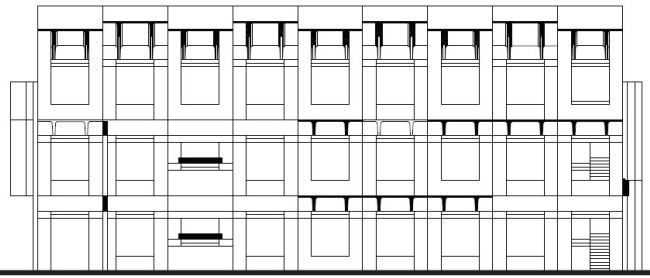




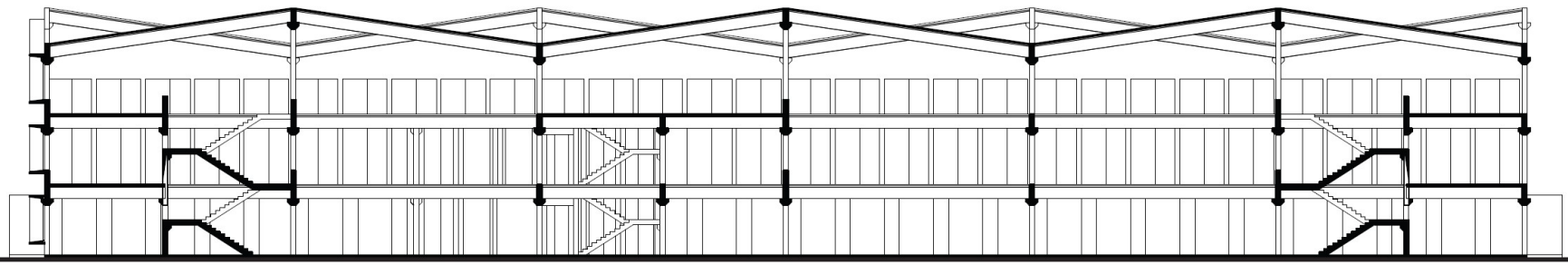
Section:
T1



Section:
T2



Section:
L1

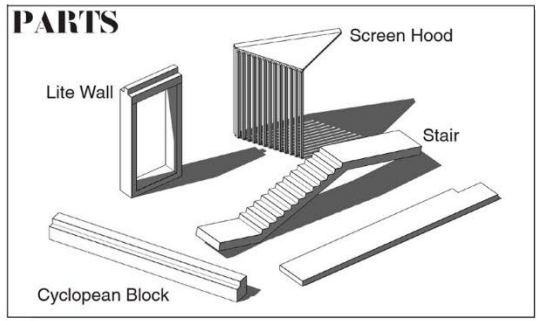


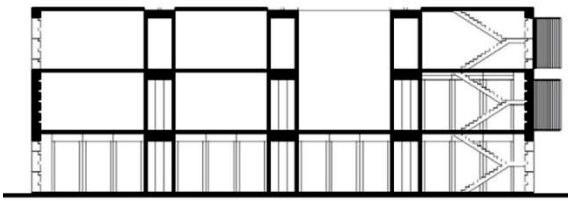
DELTA

KILO
TYPE

K.3
VARIANT





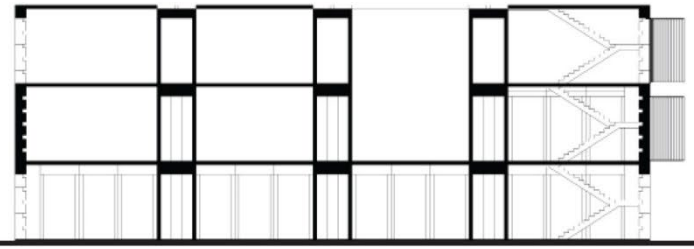




Section:
T1



Section:
T2



Section:
L1



KILO

CONCRETE BUILDING project

PROJECT-APPLIED CONCEPTUAL PARAMETERS

***…Economy** of material and optimization of structural elements is the prime driver*

***…Functional** building size, structural bays, sloped roof, daylight apertures*

***…Aesthetics** guided by Economy, Function, and the "natural" aesthetic of material and structural elements*

CONCRETE BUILDING project

PROJECT PARAMETERS: Use, Size, and Structural Loads

ENGINEERING OFFICE: Ypsilanti, Michigan (60’ x 120’ nominal footprint) (50” thick mat slab raft foundation)
ROOF LOADS: Snow Load: 20 psf Roofing DL: 15 psf
FLOOR LOADS: Floor LL: 50 psf Floor 15: 15 psf

GROCERY STORE: Pinconning, Michigan (60’ x 120’ nominal footprint) (50” thick mat slab raft foundation)
ROOF LOADS: Snow Load: 40 psf Roofing DL: 15 psf
FLOOR LOADS: Floor LL: 75 psf Floor DL: 20 psf

PUBLIC LIBRARY: Houghton, Michigan (60’ x 120’ nominal footprint) (68” thick mat slab raft foundation)
ROOF LOADS: Snow Load: 100 psf Roofing DL: 15 psf
FLOOR LOADS: Floor LL: 150 psf Floor DL: 15 psf

CONCRETE BUILDING project

PROJECT PARAMETERS: For all buildings/all students

ROOF SLOPE & STRUCTURAL DAYLIGHT OPENINGS: Slope with pitch range of 3 degrees minimum up to 10 degrees maximum.
(use only single slope pitch, not double slope i.e. pitching with two slopes in one plane)
Structural openings in roof plane for daylight: minimum of 3, maximum of 10, to work with the logic & order of the roof deck structure

HEIGHTS: Main Floor to 2nd Floor: 14'-0", and 2nd Floor to Roof: 14'-0" to 21'-0"

FLOORS: 2nd Floor with Precast Structure with 2" concrete topping; Main Floor 2" concrete topping on Mat Slab Raft Foundation

TWO STRUCTURAL PRECAST STAIRS: to work with the logic & order of the floor structure, minimum width of 48" per flight and at landing

ONE ELEVATOR SHAFT OPENING: to work with the logic & order of the floor structure, nominal 100 sf to 120 sf, and no less than 8' side

DOUBLE-HEIGHT SPACE(S): with open structural bays in 2nd Floor plan area, not to exceed 25% of 2nd Floor area

STRUCTURAL FOUNDATION: Mat Slab Raft Foundation (to resist high water-table uplift forces)

LATERAL-RESISTANT MECHANISM: Vertical Shear Walls in both directions & four sides + Horizontal Diaphragm at Floor & Wall

COLOR-CODED STRUCTURE ELEMENTS: with each element type a coded color (i.e. four elements = four colors)

CONCRETE BUILDING project

PROJECT PARAMETERS: ALPHA

ALPHA

Primary Vertical: full-height Solid and/or Lite-Wall Panels 8’ wide, and one longitudinal grid-line of Columns supporting Beams. Stair span support by Solid Panels as needed.
Primary Horizontal: Double Tees 8’ feet wide with 30’ span. Beams (L, T, or R) with 24’ span to support Double Tees. Precast Stairs/Landings with 7” R & 11” T
Secondary Lateral: Concrete Shear Wall panels and/or Lite-Walls
Foundation: Mat slab/Raft Foundation

CONCRETE BUILDING project

PROJECT PARAMETERS: CHARLIE

CHARLIE

Primary Vertical: full-height Columns to accept Beams, and one, two, or three structural grid-lines of Lite-Wall Panels.
Stair span support by Solid Panels as needed.
Primary Horizontal: Beams (L, T, or R) with 20’ span to support Hollowcore Plank; Precast Stairs/Landings with 7” R & 11” T
Secondary Horizontal: Hollowcore Plank 4’ wide with 20’ span *or* 24’ span *or* 30’ span
Secondary Lateral: Concrete Shear Wall panels and/or Lite-Walls
Foundation: Mat slab/Raft Foundation

CONCRETE BUILDING project

PROJECT PARAMETERS: DELTA

DELTA

Primary Vertical: full-height Lite-Wall Panels with option of some Solid Bearing-Wall Panels, and one transverse grid-line of Columns supporting Beams. Stair span support by Solid Panels as needed.
Primary Horizontal: Double Tees 8’ *or* 10’ feet wide with 40’ span; Precast Stairs/Landings with 7” R & 11” T
Secondary Lateral: Concrete Shear Wall panels and/or Lite-Walls
Foundation: Mat slab/Raft Foundation

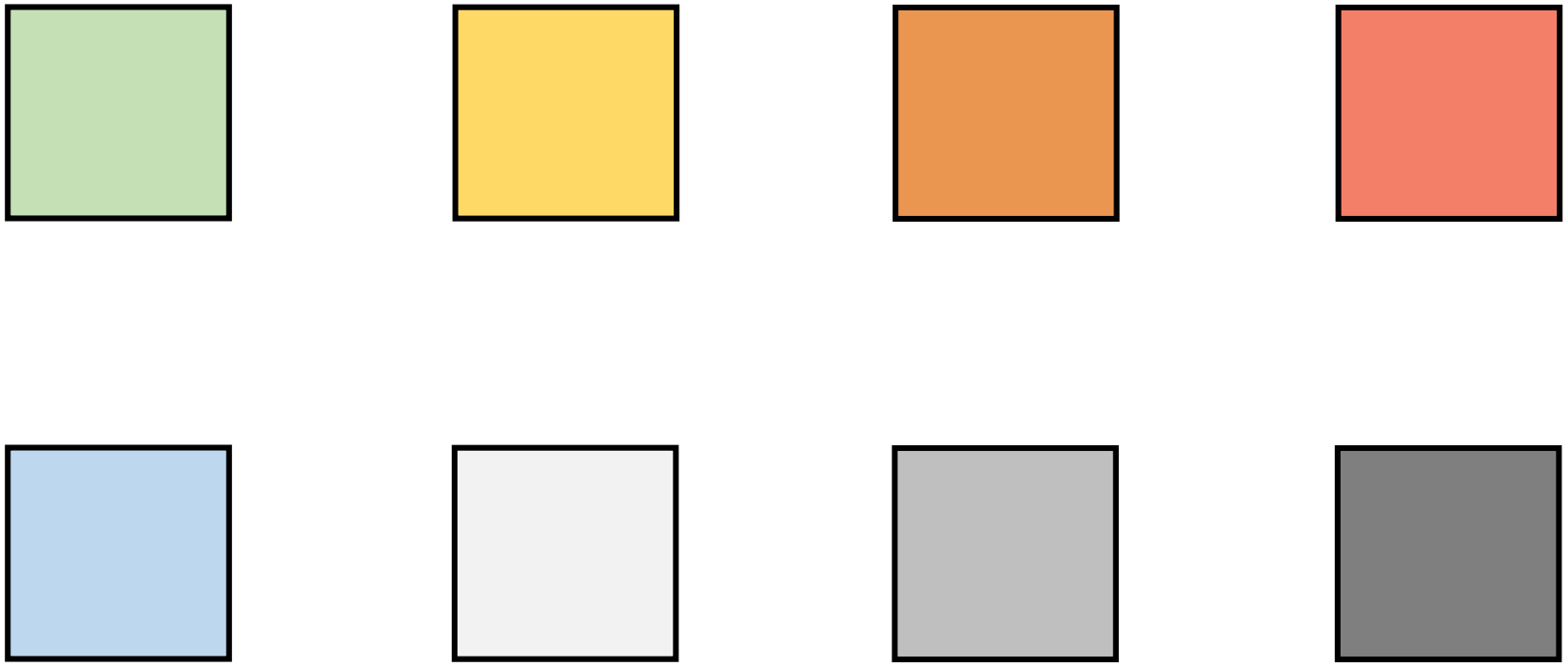
CONCRETE BUILDING project

PROJECT PARAMETERS: KILO

KILO

Primary Vertical: full-height Lite-Wall Panels to support Beams, with one longitudinal grid-line of continuous Lite-Wall to support Hollowcore. Stair spans supported by Solid Panels as needed.
Primary Horizontal: Beams (L, T, or R) with 24’ span to support Hollowcore Plank; Precast Stairs/Landings with 7” R & 11” T
Secondary Horizontal: Hollowcore Plank 4’ wide with 30’ span
Secondary Lateral: Concrete Shear Wall panels and/or Lite-Walls
Foundation: Mat slab/Raft Foundation

CONCRETE BUILDING color options for element coding

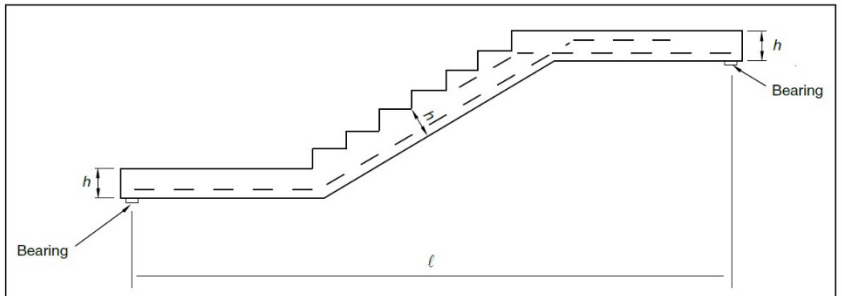




SHEAR WALLS: Transverse in foreground, longitudinal in background

PRECAST STAIR (100 psf load assumed, thickness based on span)

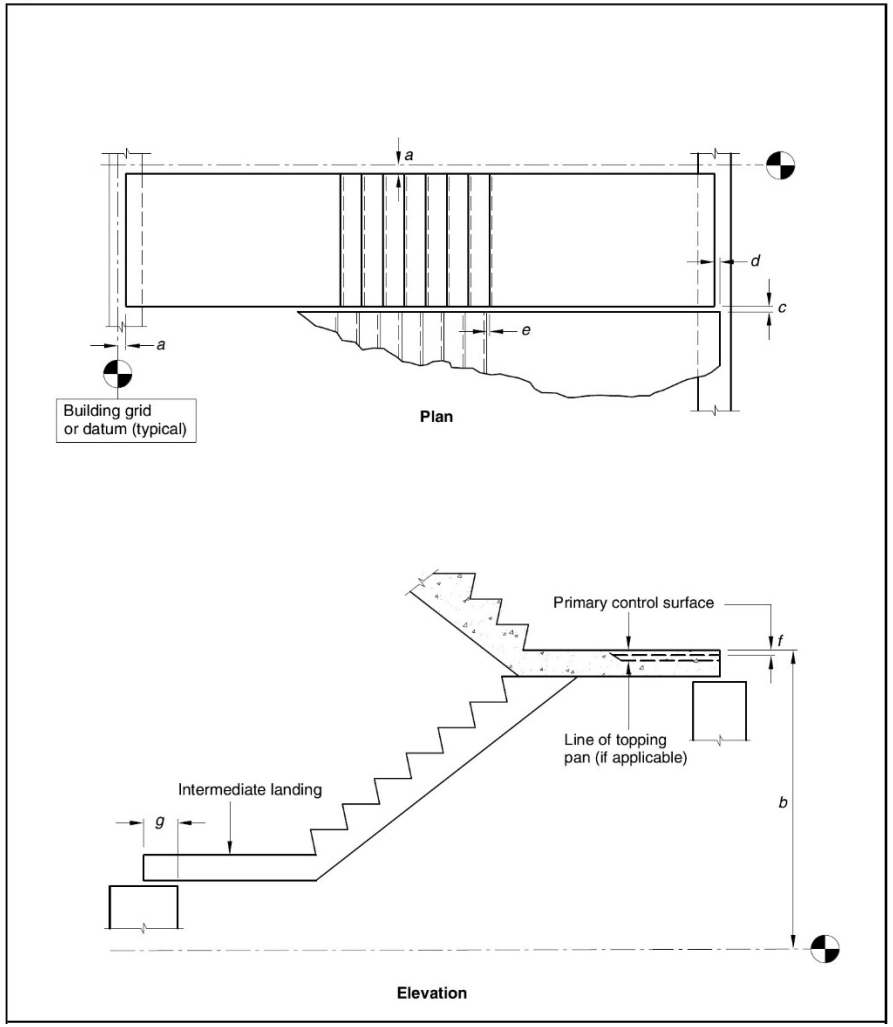
Design Aid 3.12.10 Recommended Maximum Spans for Precast Concrete Stairs



Thickness h, in.	Span ℓ, ft
6	14
8	19
10	23
12	26

Notes:
1. Stair reinforcing must be designed for all factored loads.
2. Normalweight concrete.
3. Recommended maximum span ℓ is based on dead weight of stair plus 100 lb/ft² live load.
4. Minimum concrete 28-day compressive strength is 5000 psi.
5. When deflections are required to be checked in accordance with ACI 318-05 Section 9.5, cracked section properties are required to be used. Top steel may be required to control deflections.
6. Longer spans can be obtained with higher concrete strength or special reinforcing.

Fig. 13.3.9 Stair unit erection tolerance





TRIBUTARY ELEMENTS

LOAD PATHS AND LOADS TO A COLUMN AND FOOTING

ROOF LOAD TO COLUMN

Roof Double-Tees (8DT12+2):
Roof LL + DL (45 psf) + DT self-weight (54 psf)
Roof Tributary Area = 15 ft x 24 ft = 360 sf
Roof Tributary Load to Beam = (99 psf) x 360 sf)
Roof Tributary Load to Beam = 35.640 Kips
Beam self-weight 250 plf x (12') x 2 = 6.0 Kips
Beam load in plf = 99 psf (15') + 250 plf = 1,735 plf
ROOF LOAD TO COLUMN = **41.640 Kips**

FLOOR LOAD TO COLUMN

Floor Double-Tees (8DT18+2):
Floor LL + DL (85 psf) + DT self-weight (59 psf)
Floor Tributary Area = 15 ft x 24 ft = 360 sf
Floor Tributary Load to Beam = (144 psf) x 360 sf)
Floor Tributary Load to Beam = 51.840 Kips
Beam self-weight 317 plf x (12') x 2 = 7.608 Kips
Beam load in plf = 144 psf (15') + 317 plf = 2,477 plf
FLOOR LOAD TO COLUMN = **59.448 Kips**

LOAD ON COLUMN

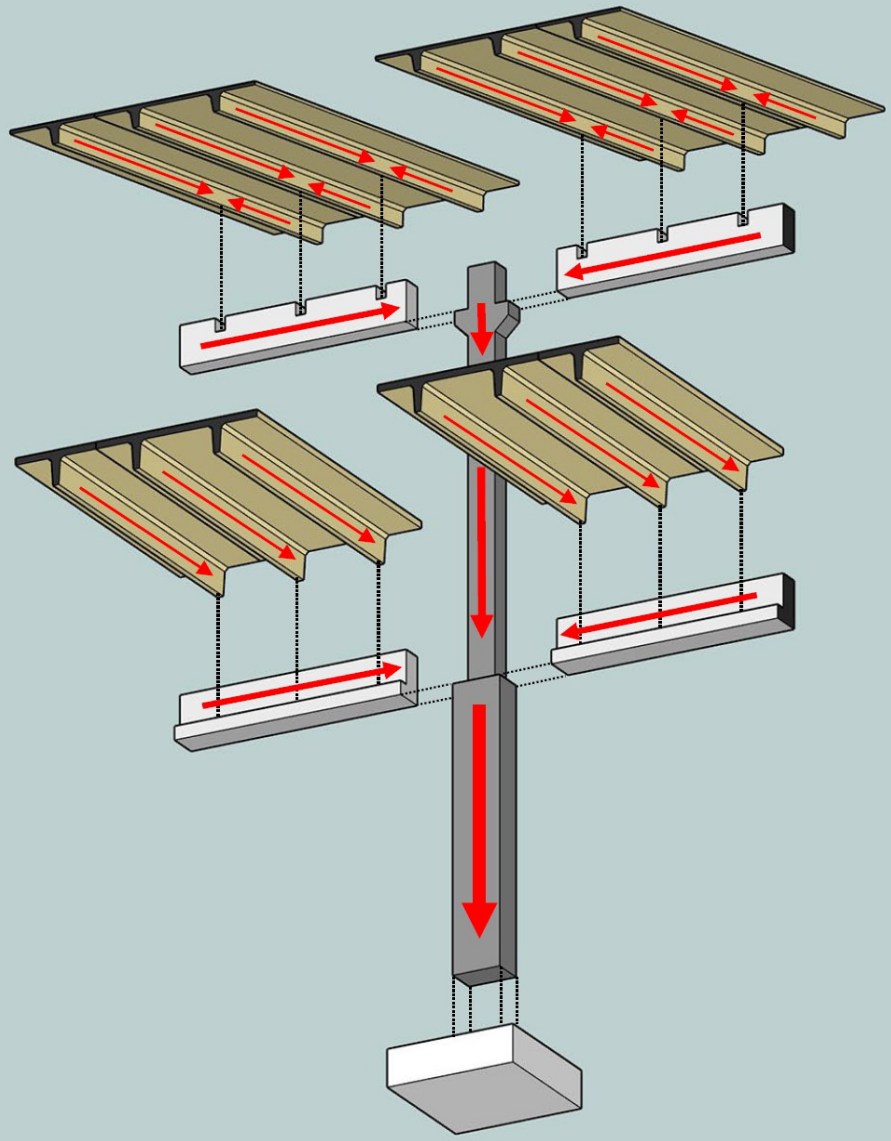
Roof 41.640 K + Floor 59.448 K = **101.088 Kips**

SELF-WEIGHT OF COLUMN

Column area x height x 150 lbs/cf concrete
Upper Column a x h = 1.78 sf x 20 ft = 35.6 cf
Upper weight = 35.6 cf x 150 lbs/cf = 5.34 Kips
Lower Column a x h = 3.5 sf x 14 ft = 49 cf
Lower weight = 49 cf x 150 lbs/cf = 7.35 Kips
COLUMN WEIGHT = 5.34 K + 7.35 K = **12.69 Kips**

LOAD ON FOOTING

Load on Column + Column self-weight
101.088 Kips + 12.69 Kips = **113.78 Kips**

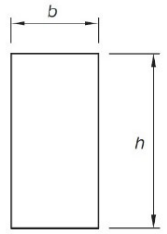


EXPLODED AXON OF TRIBUTARY ELEMENTS TRANSFERRING TRIBUTARY LOAD TO A COLUMN AND FOOTING

ROOF BEAM LOAD

12RB20 spans 24' and supports 2,620 plf
Roof Rectangular Beam load 1,735 plf < 2,620 plf allowable

3.9 Rectangular Beam Load Tables



f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter, low-relaxation strand

Normalweight concrete

Section Properties							
Designation	b in.	h in.	A in.²	I in.⁴	y_b in.	S in.³	wt lb/ft
12RB16	12	16	192	4096	8	512	200
12RB20	12	20	240	8000	10	800	250

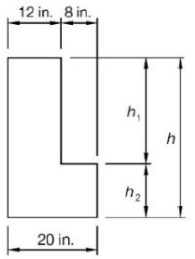
Table of superimposed service load capacity, lb/ft, and cambers, in.

Designation	Number strand	y_s in.	Span, ft									
			16	18	20	22	24	26	28	30	32	34
12RB16	5	3.00	3550	2770	2210	1790	[illegible]	1200	1040			
			0.4	0.5	0.6	0.8	0.9	1.0	1.1			
			0.2	0.2	0.2	0.2	0.3	0.3	0.3			
12RB20	8	3.00	6160	4820	3860	3150	2620	2200	1860	1600	1380	1190
			0.4	0.5	0.6	0.7	[illegible]	1.0	1.1	1.3	1.4	1.5
			0.2	0.2	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.5

FLOOR BEAM LOAD

20LB20 spans 24' and supports 2,760 plf
Floor L-Beam load 2,477 plf < 2,760 plf allowable

3.10 L-Beam Load Tables



f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter, low-relaxation strand

Normalweight concrete

Section Properties								
Designation	h in.	h_1/h_2 in.	A in.²	I in.⁴	y_b in.	S_b in.³	S_t in.³	wt lb/ft
20LB20	20	12/8	304	10,160	8.74	1163	902	317

Table of superimposed service load capacity, lb/ft, and cambers, in.

Designation	Number strand	y_s in.	Span, ft										
			16	18	20	22	24	26	28	30	32	34	36
20LB20	9	2.44	6560	5130	4100	3340	2760	2310	1960	1670	1430	1240	1070
			0.3	0.4	0.5	0.6	[illegible]	0.8	0.9	1.0	1.0	1.1	1.2
			0.1	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.2